Population size (N) = 4, Sample size (n) = 2

Population mean (μ) = (4 + 8 + 12+ 16)/4 = 40/4 = 10

The population variance is calculated as

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(X_i - \mu)^2$$

$$= \frac{1}{4}\left[(4-10)^2 + (8-10)^2 + (12-10)^2 + (16-10)^2\right] = \frac{1}{4}[36+4+4+36] = 20.$$

Hence, $SE(\bar{X}) = \sqrt{\frac{\sigma^2}{n}} = \sqrt{\frac{20}{2}} = \sqrt{10}$

This can also be verified from the sampling distribution of $\bar{X}$ (*see* Table 1.3)

$$V(\bar{X}) = \sum(\bar{x} - \mu)^2 P(\bar{X} = \bar{x})$$

where the summation is taken over all values of $\bar{x}$

Thus, $V(\bar{X}) = (4-10)^2\frac{1}{16} + (6-10)^2\frac{2}{16} + (8-10)^2\frac{3}{16} + (10-10)^2\frac{4}{16}$

$$+(12-10)^2\frac{3}{16} + (14-10)^2\frac{2}{16} + (16-10)^2\frac{1}{16}$$

$$= \frac{1}{16}(36+32+12+0+12+32+36) = 10$$

Hence, the standard deviation of the sampling distribution of $\bar{X}$ is = $\sqrt{10}$.

Standard Errors of some of the frequently referred statistics are listed in Table 1.4.

Table 1.4 Statistics and their Standard Errors

Statistic	Standard error
Sample proportion: p	$\sqrt{\frac{PQ}{n}}$, where P is the population proportion and $Q = 1 - P$.
Difference between the means $\bar{X}$ and $\bar{Y}$ of two independent samples: $(\bar{X} - \bar{Y})$	$\sqrt{\frac{\sigma_X^2}{m} + \frac{\sigma_Y^2}{n}}$ where m and n are the sizes of samples drawn from the populations whose variances are σ_X^2 and σ_Y^2 respectively.
	$\sigma\sqrt{\frac{1}{m} + \frac{1}{n}}$, where σ^2 is the common variance of the populations.
Difference between the proportions p_X and p_Y of two independent samples: $(p_X - p_Y)$	$\sqrt{\frac{P_X Q_X}{m} + \frac{P_Y Q_Y}{n}}$, where m and n are sizes of the samples drawn from the populations whose proportions are respectively P_X and P_Y; $Q_X = 1 - P_X$, $Q_Y = 1 - P_Y$.
	$\sqrt{\hat{p}\hat{q}\left(\frac{1}{m} + \frac{1}{n}\right)}$, where $\hat{p} = \frac{mp_X + np_Y}{m+n}$, $\hat{q} = 1 - \hat{p}$, m and n are sample sizes, when P_X and P_Y are unknown.

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